

**Table 1:** Key Units

|                 |                                                     |
|-----------------|-----------------------------------------------------|
| $N_{\text{av}}$ | $6.022\,14 \times 10^{23} \text{ mol}^{-1}$         |
| $R$             | $8.314\,472 \text{ J mol}^{-1} \text{ K}^{-1}$      |
| $R$             | $8.2057 \times 10^{-2} \text{ L atm/mol/K}$         |
| $k_B$           | $1.380\,65 \times 10^{-23} \text{ J K}^{-1}$        |
| $k_B$           | $8.617\,34 \times 10^{-5} \text{ eV K}^{-1}$        |
| $h$             | $6.626\,07 \times 10^{-34} \text{ J s}$             |
| $k_B/h$         | $20\,836\,619\,123.3 \text{ K}^{-1} \text{ s}^{-1}$ |
| $c$             | $2.997\,924\,58 \times 10^8 \text{ m s}^{-1}$       |

**Table 2:** Stoichiometry Equations

|                   | one reaction                                | reaction network                             |
|-------------------|---------------------------------------------|----------------------------------------------|
| balanced reaction | $\sum_j \nu_j A_j = 0$                      | $\sum_j \nu_{ij} A_j = 0, \forall i$         |
| advancement       | $n_j = n_{j0} + \nu_j \xi$                  | $n_j = n_{j0} + \sum_i \nu_{ij} \xi_i$       |
| in flow           | $F_j = F_{j0} + \nu_j \dot{\xi}$            | $F_j = F_{j0} + \sum_i \nu_{ij} \dot{\xi}_i$ |
| mole fraction     | $y_j = n_j / \sum_j n_j = F_j / \sum_j F_j$ |                                              |
| partial pressure  | $P_j = y_j P$                               |                                              |
| concentration     | $C_j = n_j / V = F_j / v$                   |                                              |
| ideal gas law     | $V = (RT/P) \sum_j n_j$                     |                                              |
| in flow           | $v = (RT/P) \sum_j F_j$                     |                                              |

**Table 3:** Activity Models

|                |                         |
|----------------|-------------------------|
| pure substance | $a_j = 1$               |
| ideal gas      | $a_j = y_j P / P^\circ$ |
| solute         | $a_j = C_j / C^\circ$   |

**Table 4:** Thermodynamic Equations

|                               |                                                                                      |
|-------------------------------|--------------------------------------------------------------------------------------|
| First Law                     | $dU = 0$ in any process in a closed system                                           |
| Second Law                    | $dS \geq 0$ in any spontaneous process in a closed system                            |
| Third Law                     | $S(0\text{ K}) = 0$                                                                  |
| Enthalpy                      | $H = U + PV$                                                                         |
| Gibbs energy                  | $G = H - TS$ minimized in any spontaneous reaction                                   |
| std reaction enthalpy         | $\Delta H_{\text{rxn}}^{\circ}(T) = \sum_j \nu_j \Delta H_{f,j}^{\circ}(T)$          |
| std reaction entropy          | $\Delta S_{\text{rxn}}^{\circ}(T) = \sum_j \nu_j S_j^{\circ}(T)$                     |
| mixture free energy           | $G(n_j, T, P) = \sum_j n_j \mu_j$                                                    |
| reaction free energy gradient | $dG = \sum_j \mu_j dn_j$                                                             |
| chemical potential            | $\mu_j = \mu_j^{\circ}(T) + RT \ln a_j$                                              |
| chemical equilibrium          | $\partial G / \partial \xi = 0$                                                      |
| mass action expression        | $Q(\xi) = \prod_j a_j^{\nu_j}$                                                       |
| equilibrium constant          | $K_{\text{eq}}(T) = \exp(-\Delta G^{\circ}(T)/RT) = \prod_j a_{j,\text{eq}}^{\nu_j}$ |
| Gibbs-Helmholtz relation      | $\partial(G/T) / \partial T = -H/T^2$                                                |
| van't Hoff approximation      | $\ln(K_{\text{eq}}(T_2)/K_{\text{eq}}(T_1)) = -(\Delta H^{\circ}/R)(1/T_2 - 1/T_1)$  |

**Table 5:** Kinetics Equations

| Arrhenius expression  | $\ln k = -E_a/RT$ |                         |                         |
|-----------------------|-------------------|-------------------------|-------------------------|
|                       | Rate              | Integrated rate         | half-life ( $t_{1/2}$ ) |
| Zeroth-order reaction | $r = k$           | $C_A = C_{A0} - kt$     | $C_{A0}/2k$             |
| First-order reaction  | $r = kC_A$        | $C_A = C_{A0}e^{-kt}$   | $\ln 2/k$               |
| Second-order reaction | $r = kC_A^2$      | $1/C_A = 1/C_{A0} + kt$ | $1/C_{A0}k$             |